DP Biology – A4.1

Evolution and speciation

- What evidence supports evolution?
- What do analogous and homologous structures show about what organisms share and how they differ?

SL statements A4.1.1–7. Textbook pp. 138–150.

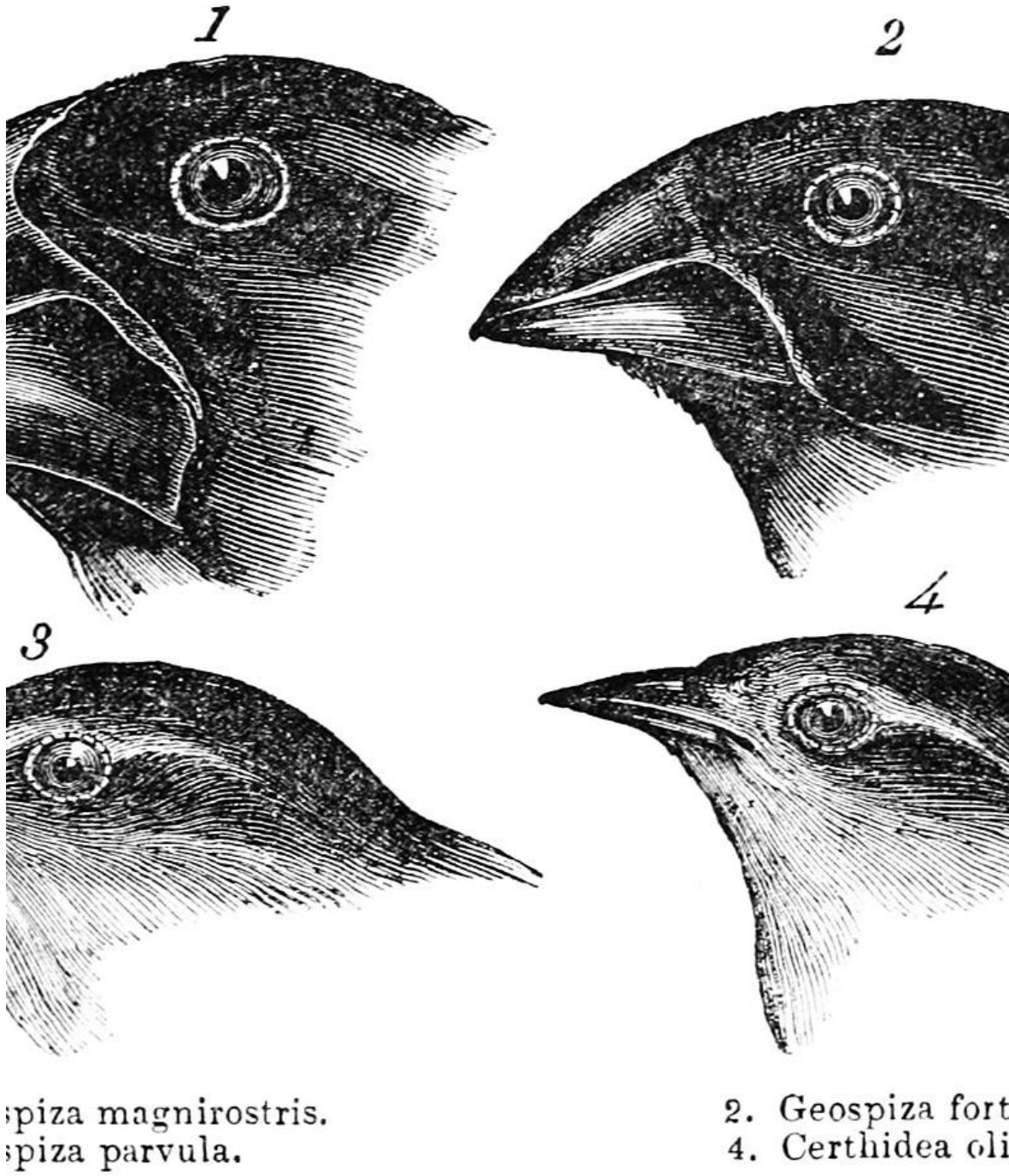


Image: Galapagos finches, drawn by John Gould. Public domain.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	A4.1.1–3	Define evolution, separate it from inheritance of acquired traits, and use sequence data and selective breeding as evidence
2	A4.1.4–5	Distinguish homologous from analogous structures, and explain convergent evolution
3	A4.1.6–7	Explain speciation, using reproductive isolation and differential selection

The big picture: change over generations

Evidence	Patterns	New species
Sequences, breeding, anatomy	Homologous vs analogous	Isolation + selection

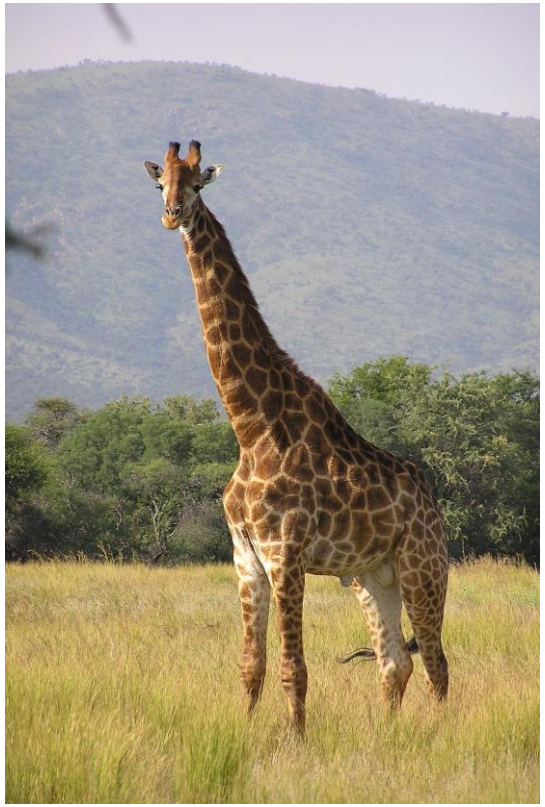
Many **independent** lines of evidence point to the same conclusion: species **change** and share **common ancestors**.

Big question
Why do bonobos and chimpanzees, so alike, never interbreed in the wild?

What is evolution?

A4.1.1

Evolution: a change in the inherited traits of a population over generations



Giraffe. M. Duchacek, CC BY-SA 3.0.

	Lamarck	Darwin
Idea	Acquired traits passed on	Selection acts on existing variation
Giraffe	Stretching lengthens necks	Longer-necked giraffes leave more young

Not evolution

Wind-bent trees grow bent, but their seedlings grow upright.

Acquired or inherited?

A4.1.1

Acquired in life: not passed on

Muscles from training; a scar; a language learned

In the genes: can be passed on

Eye colour; lactose tolerance; beak shape in finches

Sort

Dark skin after sunbathing; a new antibiotic-resistance mutation; a dyed fleece; long legs. Which can evolve?

Evidence from sequences

A4.1.2

Mutations build up over time, so close relatives have **more similar** DNA and protein sequences.

Species	Made-up section of a gene
1	A A T C G C C G A A G C T T T G C A T T T G C A A
2	A A T C G C A G A A G C T T T G C A T T T T C A A
3	A A T C A A C G A A G C T A T G C A T T T G C A A
4	A A T C A C C G A A G C T A T G C A T T T G C A A

Activity

Count differences for each pair. Which two are most closely related?

Why sequence evidence is so strong

A4.1.2

Universal

One genetic code; 20 amino acids

Shared genes

Pax6 builds eyes in flies and humans

Countable

Differences give objective data

Key idea

Sequence evidence agrees with anatomy and fossils. Independent lines of evidence pointing the same way is why scientists accept evolution.

Working with data: comparing a protein

A4.1.2

Amino acids differing in the haemoglobin beta chain, compared with humans (published values).

Species	Differences
Gorilla	1
Rhesus monkey	8
Dog	15
Frog	67

1. Rank by relatedness to humans.
2. Explain how these data support common ancestry.
3. Give one limitation of one protein.

Evidence from selective breeding

A4.1.3



Teosinte and maize. John Doebley, CC BY 3.0.

Humans choose which individuals breed. In a few thousand years, varieties become **very different** from wild ancestors.

Key idea

It shows how fast selection can change heritable traits, though it does not prove natural evolution.

One wild plant, many crops

A4.1.3



Wild cabbage. MPF, CC BY 2.5.

Cabbage	Kale	Broccoli	Cauliflower	Kohlrabi
selected for leaf buds	selected for leaves	selected for flower stalks	selected for flower buds	selected for stem

Activity

Pick a domesticated species and list three traits changed from its wild relative.

Check-in: class 1

A4.1.1–3

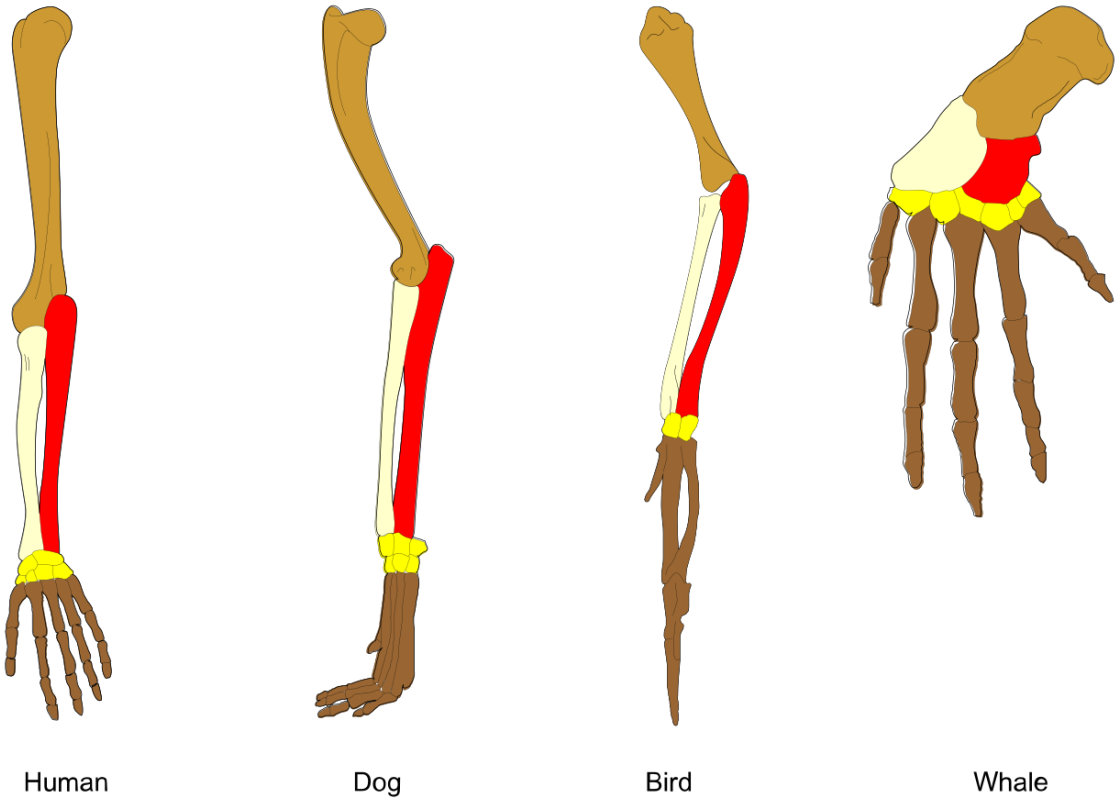
1. Define evolution.
2. Explain why acquired traits are not evolution.
3. Explain why sequence differences show relatedness.
4. Explain what selective breeding shows.
5. Give an example of selective breeding.

How to use this

Answer without your notes, then check against slides 3–10.

Evidence from homologous structures

A4.1.4



Homologous limb bones. Volkov Vladislav Petrovich, CC BY-SA 4.0.

Homologous structures share a basic plan but may have different functions.

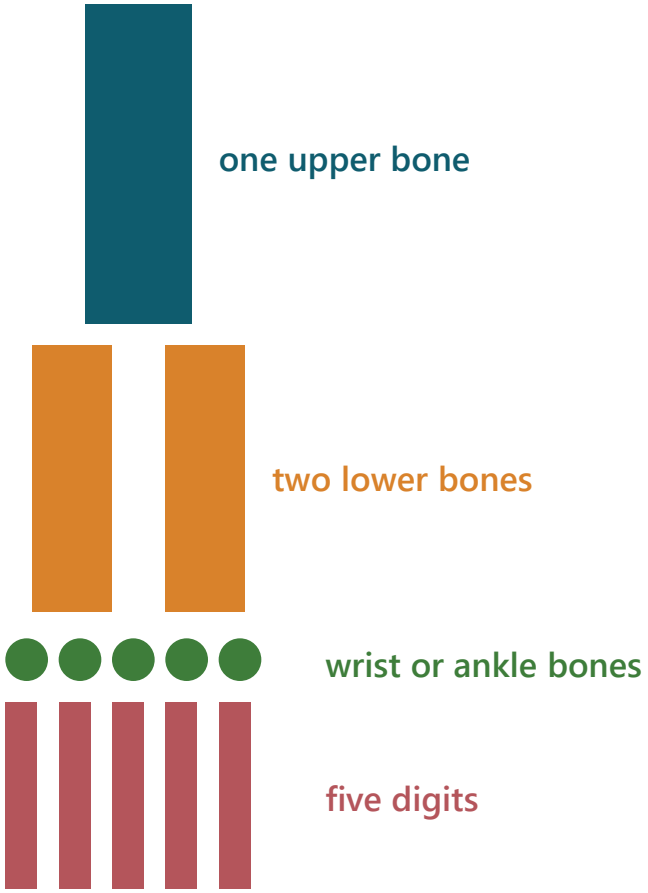
They suggest a **common ancestor**: **divergent** evolution.

Example

Whale flippers, bat wings and human arms share one bone pattern.

The pentadactyl plan

A4.1.4



Use it

Colour each bone type on a bat, whale, horse and human limb. Why would the same plan appear in all if they did not share an ancestor?

The pentadactyl limb

A4.1.4

Region	Forelimb	Hindlimb
One upper bone	Humerus	Femur
Two lower bones	Radius and ulna	Tibia and fibula
Wrist or ankle	Carpals	Tarsals
Digits	Metacarpals and phalanges	Metatarsals and phalanges

Adaptations

Bat: long digits for wings. Whale: short, flat paddle. Horse: one large digit for running.

Convergent evolution and analogous structures

A4.1.5

Analogous structures do the same job but have different origins. They arise by **convergent evolution**: similar pressures on unrelated groups.

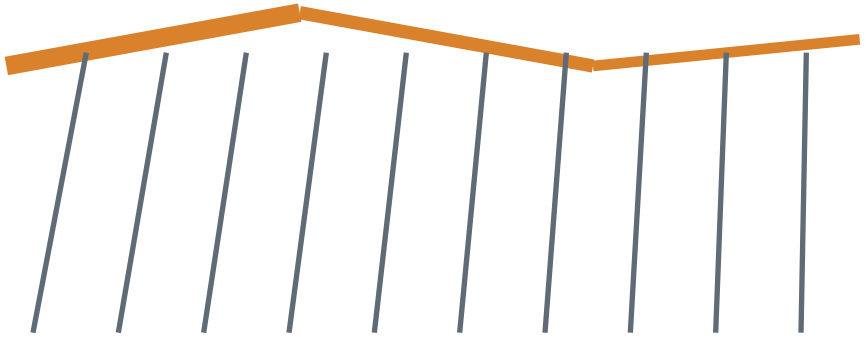
	Homologous	Analogous
Basic structure	Similar	Different
Function	May differ	Similar
Ancestry	Shared	Separate
Example	Bat wing, whale flipper	Bird wing, insect wing

Key idea

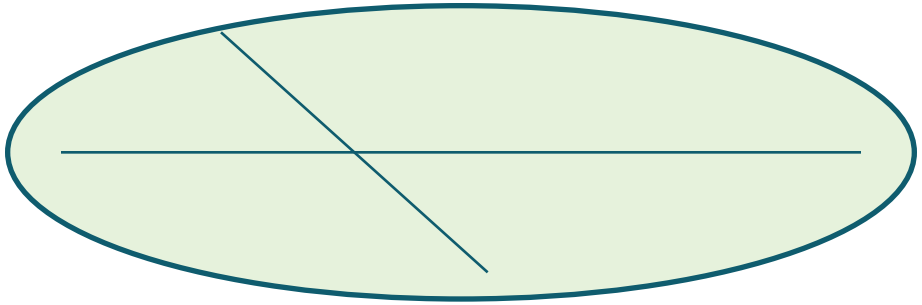
Appearance can mislead; DNA sequences now guide classification.

Two wings, two origins

A4.1.5



bird: bones of a pentadactyl limb + feathers



insect: thin chitin membrane, no bones

So

Same function (flight), different structure and origin: analogous. Natural selection for flight produced similar shapes twice.

Practice: homologous or analogous?

A4.1.4–5

Classify

Shark fin and dolphin flipper.

Classify

Human arm and penguin flipper.

Classify

Octopus eye and human eye.

Explain

Give the reason for each answer, using structure and ancestry.

Check-in: class 2

A4.1.4–5

1. Define homologous structures.
2. Describe the pentadactyl limb.
3. Define analogous structures.
4. Explain convergent evolution.
5. Explain why appearance can mislead classification.

How to use this

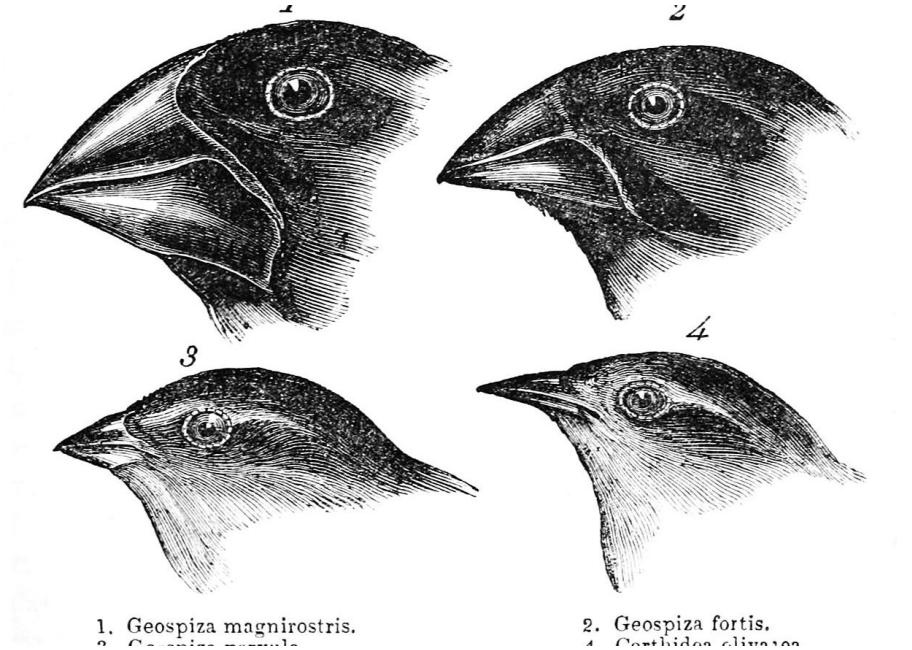
Answer without your notes, then check against slides 12–17.

Speciation

A4.1.6

Speciation is the splitting of one species into two or more: the only way new species arise.

Speciation adds species; extinction removes them.



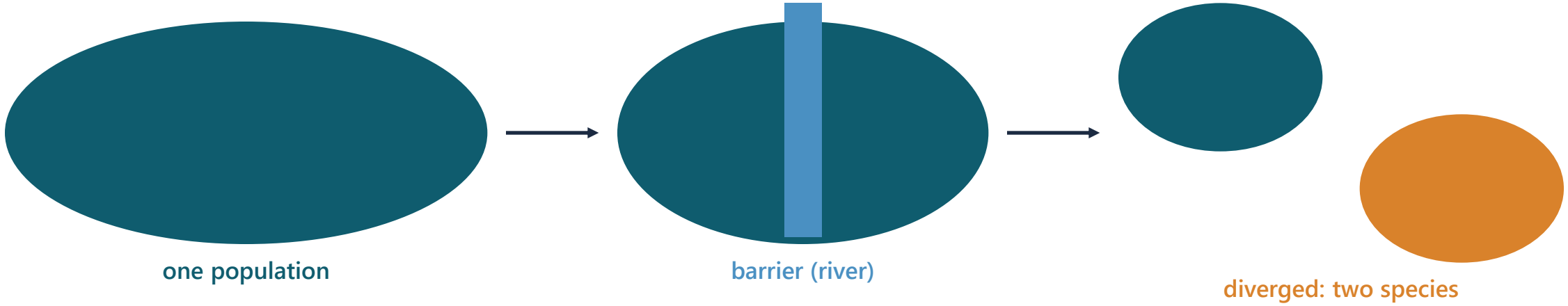
Galapagos finches. John Gould, public domain.

Example

Galapagos finches probably descend from one ancestor; populations adapted to different foods and beaks diverged.

How speciation happens

A4.1.7



Requirement	Meaning
Reproductive isolation	No gene flow between populations
Differential selection	Different pressures make traits diverge

Common mistake

Isolation alone is not enough; selection must make them diverge.

Geographical isolation

A4.1.7



Grand Canyon, which separates squirrel populations. Chensiyuan, CC BY-SA 4.0.

Rivers, mountains or canyons can split a population.

With no gene flow, mutations and selection build differences until the populations can no longer interbreed.

Think

Why might a canyon isolate squirrels but not birds?

Working with data: island lizards

A4.1.7

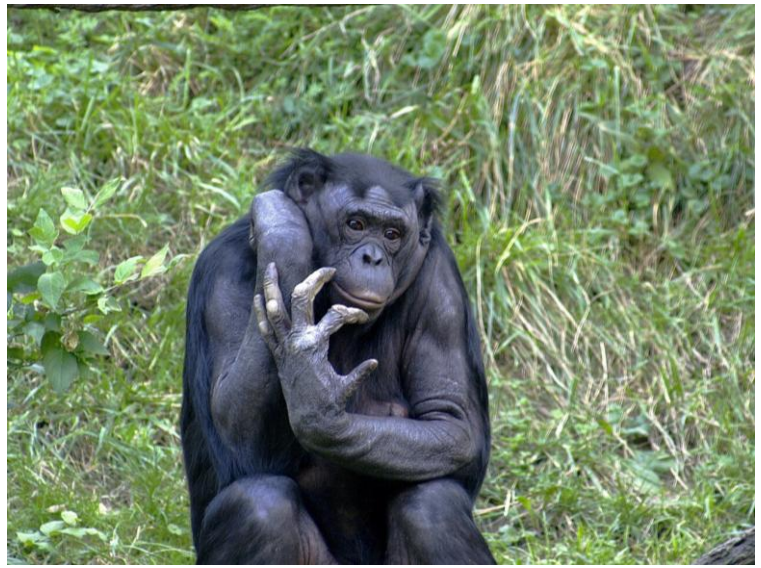
Mean hind-leg length in lizards on two islands, 30 years after a few were moved from island A to island B. Invented values; realistic pattern.

Year	Island A / mm	Island B / mm
0	25	25
10	25	23
20	25	21
30	25	20

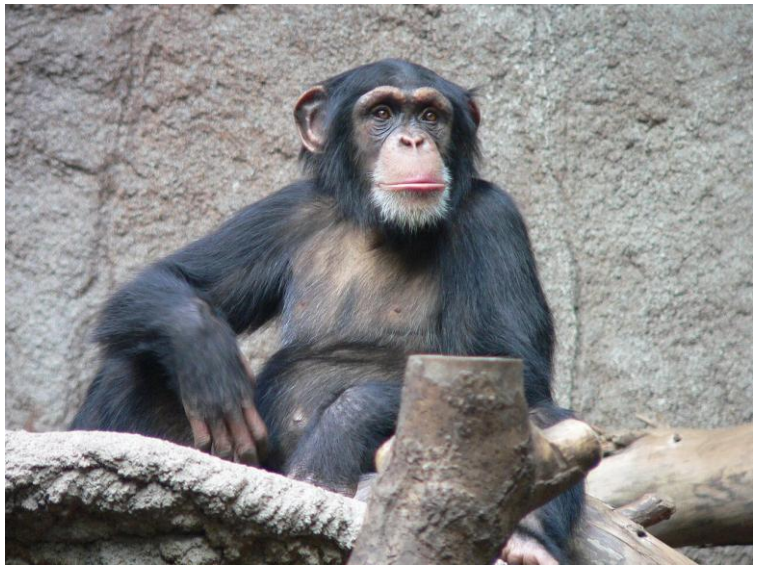
1. Describe the change on island B.
2. Suggest a selection pressure that could cause it.
3. What would show the populations are separate species?

Example: bonobos and chimpanzees

A4.1.7



Bonobo. CC BY-SA 2.5.



Chimpanzee. Thomas Lersch, CC BY 2.5.

Separated by the **Congo River**; neither swims.

Big question

South of the river, bonobos faced less competition for food; diet and social behaviour diverged. Females lead bonobo groups; chimpanzee groups are male-led.

Practice: speciation

A4.1.6–7

Outline

How one population becomes two species.

Explain

Why both isolation and selection are needed.

Apply

The bonobo and chimpanzee example.

Use

reproductive isolation, gene flow, differential selection, diverge

Check-in: class 3

A4.1.6–7

1. Define speciation.
2. State the two requirements for speciation.
3. Explain how a river can cause speciation.
4. Explain why isolation alone is not enough.
5. Outline the bonobo and chimpanzee example.

How to use this

Answer without your notes, then check against slides 19–24.

Common mistakes to avoid

Not this	But this
Individuals evolve in their lifetime.	Populations evolve over generations.
Humans evolved from chimpanzees.	They share a common ancestor.
Similar structures always mean close relatives.	Analogous structures arise by convergence.
Isolation alone makes a new species.	Selection must also act.

Summary: evolution and speciation

A4.1.1–7

	Key idea
A4.1.1	Evolution: inherited traits change over generations
A4.1.2	Sequence evidence
A4.1.3	Selective breeding
A4.1.4	Homologous structures; pentadactyl limb
A4.1.5	Analogous structures; convergence
A4.1.6–7	Speciation: isolation + differential selection

Check that you can...

1. **Define** evolution
2. **Evaluate** sequence evidence
3. **Explain** selective breeding as evidence
4. **Distinguish** homologous and analogous structures
5. **Outline** speciation

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Define evolution.
2. Distinguish between homologous and analogous structures, with one example of each.
3. State the two processes required for speciation.

Image credits

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